Review 1)

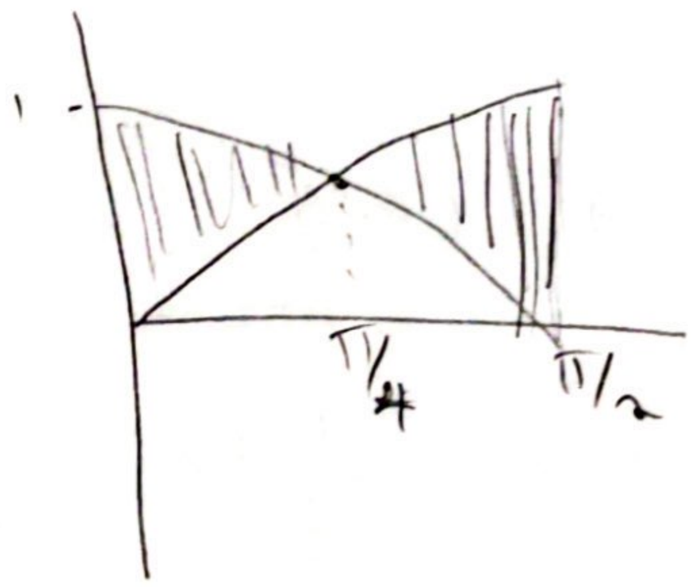
1 Find the area of the region enclosed by

a) $y = \sin x$, $y = \cos x$, $x = 0$, $x = \pi/2$.

Sketch the graphs.

- $\sin x = \cos x$

$\Rightarrow x = \pi/4$



$$\int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx$$

$$= \sin x + \cos x \Big|_0^{\pi/4} + (-\cos x - \sin x)\Big|_{\pi/4}^{\pi/2}$$

$$= (\sqrt{2} - 1) + (-1 + \sqrt{2})$$

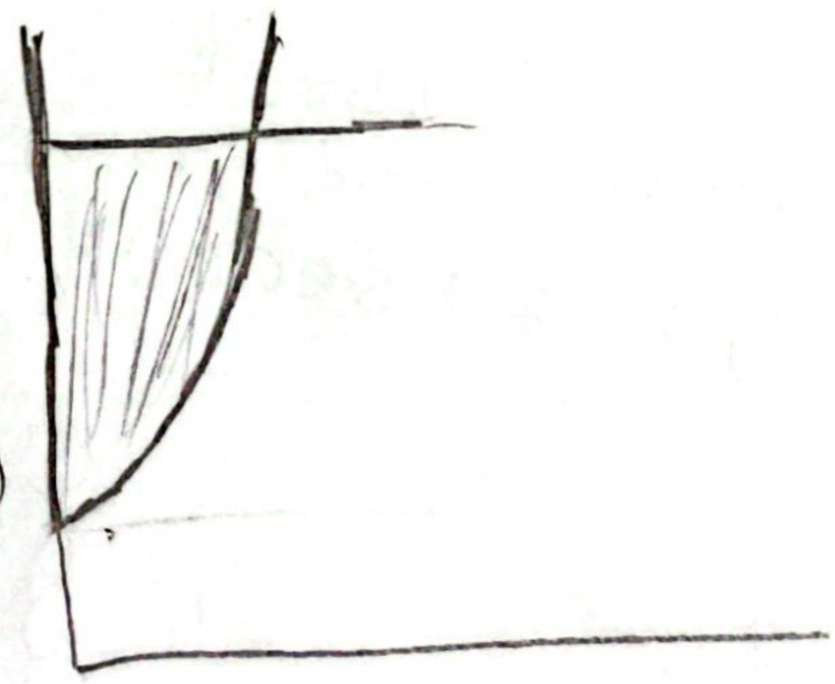
$$= \boxed{2\sqrt{2} - 2}$$

b) $y = e^{3x}$, $y = 3$ & $x = 0$

$e^{3x} = 3$

$3x = \ln(3)$

$x = \ln(3)/3$



$$\int_0^{\ln(3)/3} (3 - e^{3x})\,dx = 3x - \frac{1}{3}e^{3x}\Big|_0^{\frac{\ln(3)}{3}}$$

$$= \boxed{\left(\ln 3 - \tfrac{2}{3}\right)}$$

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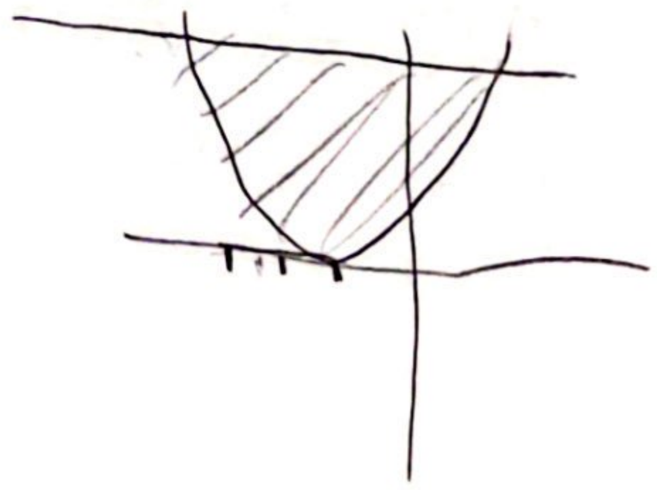
c) $y=(x+1)^2$, $y=4$

$(x+1)^2=4$

$x+1=\pm 2$

$x=-3,1$



$$\int_{-3}^{1}\left(4-(x+1)^2\right)dx$$

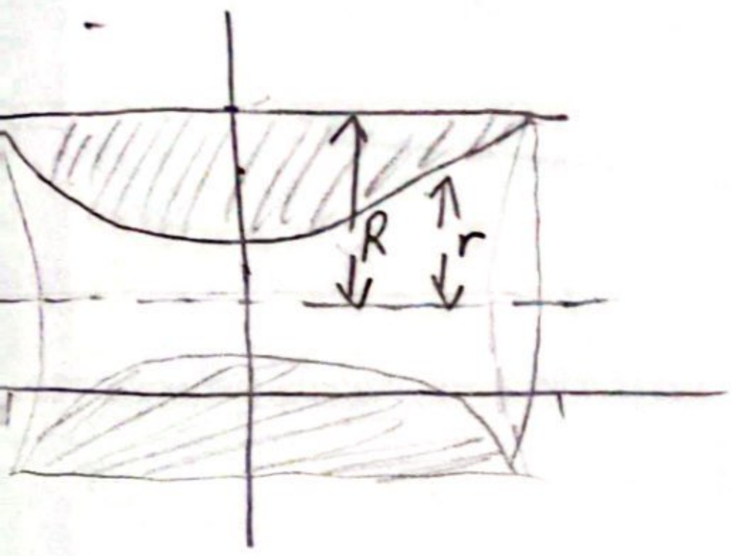
$$=4x-\frac{1}{3}(x+1)^3\Big|_{-3}^{1}$$

$$=\left(4(1)-\frac{1}{3}((1)+1)^3\right)-\left(4(-3)-\frac{1}{3}((-3)+1)^3\right)$$

$$=\boxed{\frac{32}{3}}$$

2) Find the volume of the solid formed rotating the regions bounded by the given curves about the specified lines sketch the graphs

a) $y=3+\sec(x)$, $y=5$, $-\pi/3 \le x \le \pi/3$ about $y=3$.



$$V=\int_{-\pi/3}^{\pi/3}\pi\left(R^2-r^2\right)dx$$

$R=5-3=2$

$r=3+\sec(x)-3=\sec x$

$$\int_{-\pi/3}^{\pi/3}\pi\left(2^2-(\sec x)^2\right)dx=\int_{-\pi/3}^{\pi/3}\pi\left(4-\sec^2 x\right)$$

$$=4x-\tan x\Big|_{-\pi/3}^{\pi/3}=\boxed{\left(\frac{8\pi}{3}-2\sqrt{3}\right)\pi}$$

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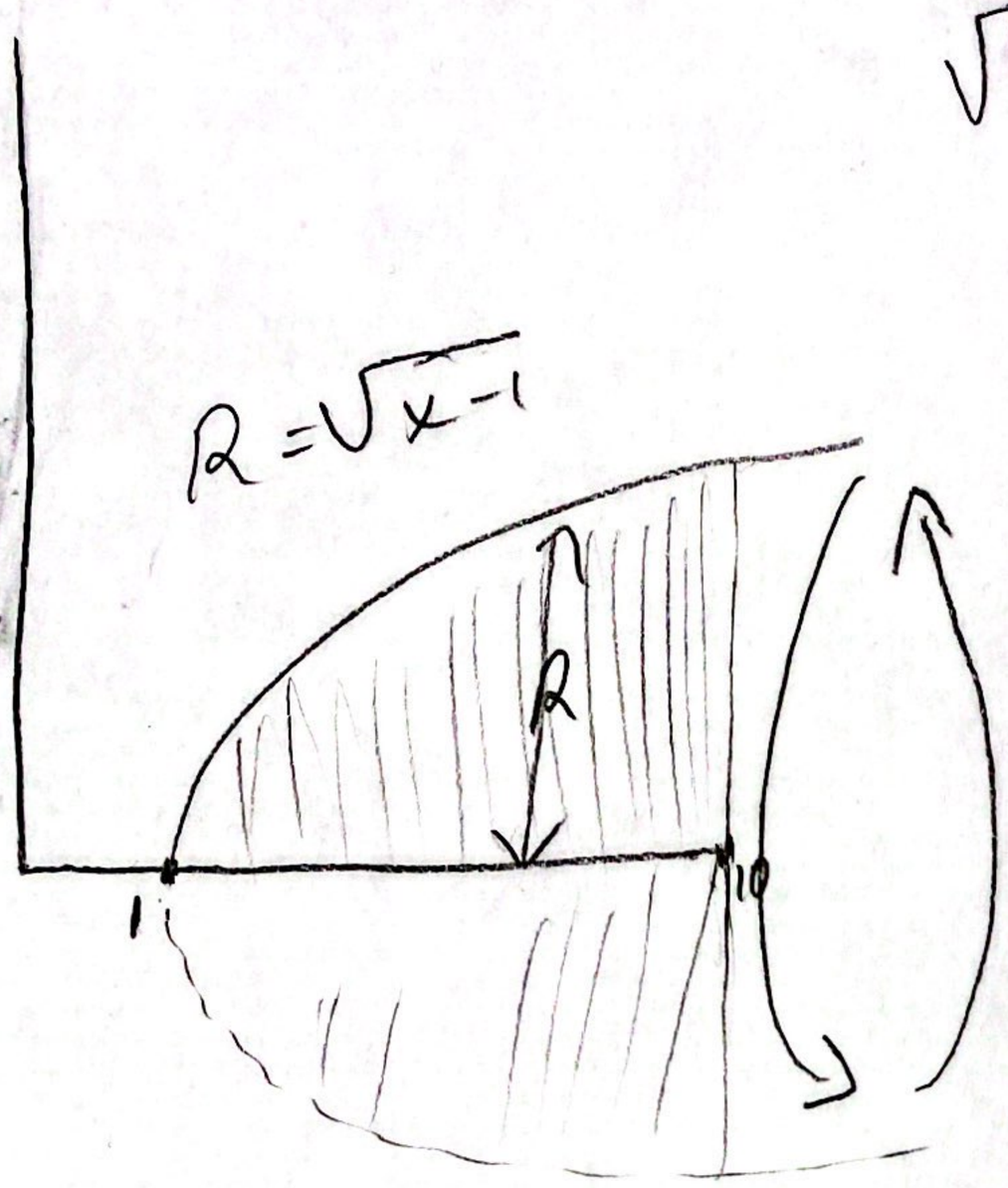
b) $y=\sqrt{x-1}$, $y=0$, $x=10$ about the x-axis

$\sqrt{x-1} = 0$

$x-1=0 \Rightarrow x=1$,



$$\int_1^{10} \pi(R^2)\,dx = \int_1^{10} \pi(\sqrt{x-1})^2\,dx$$

$$= \pi\left(\frac{x^2}{2} - x\right)\Big|_1^{10}$$

$$= \pi\left(\frac{10^2}{2} - 10\right) - \left(-\pi/2\right)$$

$$= \boxed{\frac{81\pi}{2}}$$

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